THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2012: Fundamental Concepts of Mathematics

May 13, 2021 9:30 a.m. – 12:00 noon

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL SIX questions

Note: You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**

Remark: Begin the answer to each question on a new page (sub-parts of a question can be put in the same page).

1. (24 points) Prove or disprove each of the following statements:

   (a) For any sets A, B, C, if $A \cup C = B \cup C$ and $A \cap C = B \cap C$, then $A = B$.

   (b) For any nonempty sets A, B, $\mathcal{P}(A \triangle B) = \mathcal{P}(A \setminus B) \cup \mathcal{P}(B \setminus A)$.

   (c) Let $A = \{n \in \mathbb{N} : n = x^2 + xy + y^2 \text{ for some } x, y \in \mathbb{N}\}$. For any $p \in A$, if $p > 3$ and p is prime, then $p \equiv 1 \pmod 6$.

   (d) There exist functions $f : \mathbb{R} \longrightarrow \mathbb{R}$ and $g : \mathbb{R} \longrightarrow \mathbb{R}$ such that

$$(f \circ g)(x) = x^2 \quad \text{and} \quad (g \circ f)(x) = x^3.$$

2. (20 points)

   (a) Prove that for any $n \in \mathbb{N}$, there exist $x_n, y_n \in \mathbb{N}$ such that

$$x_n + y_n\sqrt{3} \;=\; (2 + \sqrt{3})^n.$$

   The remaining parts of this question will be about the sequence $x_1, x_2, x_3 \ldots$ of numbers given in (a).

   (b) Prove that for any $n \in \mathbb{N}$, if n is odd, then x_n is even.

   (c) Prove that for any $n \in \mathbb{N}$, if $n \equiv 0 \pmod 4$, then $x_n \equiv 1 \pmod 4$.

   (d) Hence, or otherwise, prove that there is exactly one number in the sequence $x_1, x_2, x_3 \ldots$ which is prime and is congruent to 3 (mod 4).

3. (12 points) Let $A = \mathbb{R}^2 \setminus \{(0,0)\}$ and let R be a relation on A defined by

$$(x_1, y_1)R(x_2, y_2) \quad \text{if } (x_1, y_1) = (\lambda x_2, \lambda y_2) \quad \text{for some } \lambda \in \mathbb{R} \setminus \{0\}.$$

   (a) Prove that R is an equivalence relation on A.

   (b) For any $(x, y) \in A$, denote by $[(x, y)]$ the equivalence class of (x, y) with respect to R. Define

$$P_1 = \{[(0,1)]\} \quad \text{and} \quad P_2 = \{[(1, m)] : m \in \mathbb{R}\}.$$

   Prove that $P_1 \cup P_2$ is a partition of A.

4. (12 points) Let $f : \mathbb{N} \longrightarrow \mathbb{N}$ be a function satisfying

$$f(f(m) + f(n)) = m + n \quad \text{for any } m, n \in \mathbb{N}. \qquad (*)$$

(a) Prove that f is injective.

(b) Prove that $f(n+1) - f(n) = f(2) - f(1)$ for any $n \in \mathbb{N}$. Hence, or otherwise, find all functions $f : \mathbb{N} \longrightarrow \mathbb{N}$ satisfying $(*)$.

5. (20 points) In this question, you may use without proof the following facts (here we denote by $\mathbb{I}$ the set of irrational numbers):

- $\sqrt{2} \in \mathbb{I}$
- $\mathbb{Q}$ and $\mathbb{I}$ are infinite subsets of $\mathbb{R}$
- $\mathbb{Q}$ is denumerable
- $\mathbb{R}$ is uncountable
- For any sets X, Y, if X and Y are denumerable, then $X \cup Y$ is also denumerable.

(a) Prove that $\mathbb{I}$ is uncountable.

(b) Prove that there exists a denumerable subset S of $\mathbb{I}$.

(c) Prove that for any nonempty sets A, B and C, if A is numerically equivalent to B and $A \cap C = B \cap C = \emptyset$, then $A \cup C$ is numerically equivalent to $B \cup C$.

(*Hint*: Construct a bijective function $g : A \cup C \longrightarrow B \cup C$. Verify that the function g you constructed is well-defined and bijective.)

(d) Use the results from (b) and (c) to prove that $\mathbb{I}$ is numerically equivalent to $\mathbb{R}$.

6. (12 points) Let A and B be nonempty sets and let $f : A \longrightarrow B$ be a function. Consider the following statements:

(1) $f(S) \setminus f(T) = f(S \setminus T)$ for any subsets S and T of A.

(2) $f^{-1}(U) \setminus f^{-1}(V) = f^{-1}(U \setminus V)$ for any subsets U and V of B.

(a) Exactly one of the above statements is true and the other is false. Give a proof for the true statement and a disproof for the false statement.

(b) The false statement will become true if one of "f is injective" or "f is surjective" is imposed on f. Make your assertion and prove that the false statement becomes true under the additional assumption.

******* **END OF PAPER** *******